

4(i)	They are an action-reaction pair or they are equal in magnitude and opposite in direction.	B1
(ii)	$Mr_1 \left(\frac{2\pi}{T} \right)^2 = 2Mr_2 \left(\frac{2\pi}{T} \right)^2$ $\frac{r_1}{r_2} = \frac{2M}{M} = 2$	B1
(iii)	Since $\frac{r_1}{r_2} = 2$ and $r_1 + r_2 = 3.0 \times 10^{12}$ $r_1 = \frac{2}{1+2} \times 3.0 \times 10^{12} = 2.0 \times 10^{12} \text{ m}$	C1 A1
(iv)	Consider forces acting on M. From Newton's 2nd law and Newton's law of universal gravitation, $\frac{G(2M)(M)}{(3.0 \times 10^{12})^2} = M(2.0 \times 10^{12}) \left(\frac{2\pi}{T} \right)^2$ $T^2 = \frac{(4\pi^2)(2.0 \times 10^{12})(3.0 \times 10^{12})^2}{(6.67 \times 10^{-11})(2 \times 2.0 \times 10^{30})} = 2.6635 \times 10^{18}$ $T = 1.63 \times 10^9 \text{ s}$	M1 A1

